

App Dev Project Report

1.Student Details

Name: Shriya Keshri

Roll Number: 25f2000962

Email: 25f2000962@ds.study.iitm.ac.in

About Me: I am a diploma student at IIT Madras BS Degree program with a strong interest in web development and backend systems. I enjoy building scalable applications and learning how real-world systems manage data and users efficiently.

2.Project Details

Project Title: Trekking Management Application

Problem Statement: To build a web-based Trekking Management Application for Admin, Trek Staff, and Users to manage treks, staff approvals, bookings, and trekking history.

Approach : Built using Flask with SQLAlchemy (SQLite) backend and Jinja2 with Bootstrap frontend, implementing role-based access for Admin, Trek Staff, and Users with separate dashboards and controlled booking and trek management workflows.

3.AI/LLM Declaration

Used ChatGPT (GPT-5) for SQLAlchemy models and API documentation with ~15–20% assistance, while all implementation, debugging, and integration were done manually.

4.Technologies and Frameworks Used

Technology	Purpose
Flask	Core backend web framework
SQLAlchemy	ORM for SQLite database

Jinja2	Template engine for dynamic HTML
Bootstrap 5	Frontend styling and responsive design
HTML5 & CSS	Frontend structure and styling
SQLite	Lightweight database for storing data

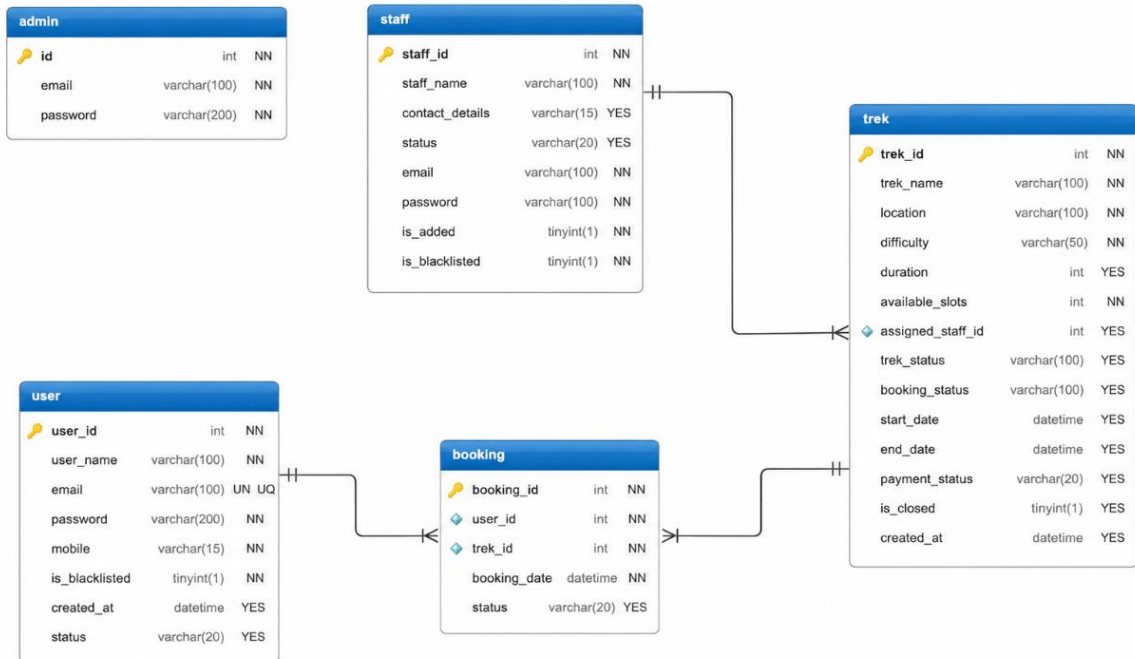
5.Database Schema / ER Diagram

Tables:

- 1.**Admin** (id, email, password)
- 2.**User** (user_id, user_name, email, password, mobile, is_blacklisted, status, created_at)
- 3.**Staff** (staff_id, staff_name, contact_details, email, password, status, is_added, is_blacklisted)
- 4.**Trek** (trek_id, trek_name, location, difficulty, duration, available_slots, assigned_staff_id, trek_status, booking_status, start_date, end_date, payment_status, is_closed, created_at)
- 5.**Booking** (booking_id, user_id, trek_id, booking_date, status)

Relationships:

- Staff** → **Trek** (One-to-Many)
- User** → **Booking** (One-to-Many)
- Trek** → **Booking** (One-to-Many)
- User** ↔ **Trek** (Many-to-Many through Booking)



6.Architecture and Features

Architecture Overview:

- app.py – main Flask application entry point
- models.py – database models using SQLAlchemy
- routes.py – Contains all Flask routes
- templates/ – Jinja2 HTML templates
- static/ – Images and background

Implemented Features:

- User and Trek Staff registration and login
- Role-based dashboards (Admin, Trek Staff, User)
- Trek Staff approval by Admin
- Trek creation and management
- Trek booking system with booking status tracking
- Admin dashboard with overall statistics
- Responsive UI using Bootstrap

Additional Features:

- Blacklisting of users and trek staff
- Prevention of overbooking
- Search functionality for Admin
- Staff assignment to treks

7. Video Presentation

Drive Link:

[Demo Video Link](#)